A user's guide to the ambiguous word 'epigenetics'

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The term 'epigenetics' is currently ambiguous, over-encompassing and uncoupled from its historical roots. This reflects interests and insights that have developed over time. In this Comment, I propose that we stop using the word in isolation, and be explicit about which definition we are using to avoid ambiguity in its scientific and public use.

The current meaning of 'epigenetics'

The evolution of the word 'epigenetics' has been excellently reviewed[1,2]. Today, the most common definition of the word is a back-translation of 'epi' (upon, above, beyond) and 'genetic' (DNA sequence), referring to a layer of information that exists beyond that encoded in the DNA sequence, thereby making the genome function distinctively in different cell types. This epi+genetics definition encompasses all chromatin and DNA modifications and other transcription regulators that act in the context of chromatin.

A historically uncoupled definition

Today's epi+genetics definition bears no resemblance to C. H. Waddington's original concept. He was describing the interaction between cell differentiation (epigenesis) and genetic mutations during development, portraying the relationship as a hillside with bifurcating canals (creodes). The choice of creode at a bifurcation indicated a cell-fate decision, with the creodes themselves formed by the actions of genes, thereby allowing a mutation to alter cell fate. As such, Waddington was describing what would become the field of molecular development.

The longer-lasting influence on the term epigenetics was its reinterpretation by D. L. Nanney in the 1950s to indicate a non-genetic 'persistent homeostasis', for the first time equating epigenetics with cellular memory. It was this redefinition to which Arthur Riggs and Robin Holliday returned in the 1970s, prompted by the then-new observations of the propagation of DNA methylation following replication. This non-sequence-dependent mechanism of transmitting information fitted nicely with a model of molecular mediation of cellular memory.

How epigenetics became molecular

The coupling of DNA methylation with epigenetics had the unintended consequence of epigenetics beginning to be equated with molecular processes. This coincided with a period of immense productivity in characterizing histone modifications and variants and with burgeoning of the field of non-coding RNAs. Paradigms of regulation of gene expression, such as in X chromosome inactivation and genomic imprinting, revealed allelic differences in chromatin organization, post-translational modifications of histones, the expression of long non-coding RNAs and other candidate transcription regulation mechanisms.

These developments supported the idea of epigenetics as molecular information that directs the same DNA sequence to act distinctively in different cell types. The definition then bifurcated like a creode: traditionalists insisted that the definition includes heritability through cell division, whereas the emerging consensus used the epi+genetics back-translation to describe broadly any information that turns the expression of genes on and off. The epi+genetics definition thus became completely dissociated from the historical roots of the term.

The problem with epi+genetics

The idea of information existing beyond the genome controlling gene expression is very attractive. The further implication that this information is malleable and can respond to perturbations and environmental influences added to the excitement about epigenetics. The vague recollection that epigenetics somehow also involved memory of past events suggested the even more intriguing idea that this epigenetic information, once perturbed, could persist and propagate the effects of temporally remote stresses.

This is where current problems occur. First, the epi+genetics definition — the layer of information switching genes on and off — is much too broad, encompassing all potential transcription regulators (but omitting transcription factors, for strange, historical reasons[3]). Second, the epi+genetics definition, with its hint of memory of past events, has led to the assumption that any change in any transcription regulator reflects cellular memory. Third, we have an immense problem interpreting differences in functional genomics assays[3].

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In essence, the issue is that a change in, for example, DNA methylation may indeed reflect changes within cells, but frequently it appears instead to be due to changes in cell subtype composition, DNA sequence variability or reverse causation (transcription altering DNA methylation rather than being altered by it)[3].

If you want to identify a change in cell state (what we have called 'cellular reprogramming' (REF. 3)), it is necessary to understand why these changes have occurred, which requires studies much more detailed than those performed to date. This lack of interpretability is a major reason why a change found in something like DNA methylation is not necessarily epigenetic.

Epigenetic properties are cellular

To help resolve the confusion in using the term epigenetics, Tuuli Lappalainen and I have proposed a return to the use of a definition based on cellular properties[3]. We define two types of events as epigenetic, the first being cellular reprogramming, the second being the formation of different proportions of cell types through the variable use of creodes ('polycreodism'). These are overtly neo-Waddingtonian descriptions of cell fates and states. Using the term epigenetics to describe cellular properties has the advantage of being testable experimentally by proposing clear hypotheses and looking for the transcription regulatory processes that mediate these events.

Unintended consequences

Our collective lack of clarity about what we mean by epigenetics has had two major unintended consequences. One stems from our scrupulous fairness in defining epigenetic events as heritable not only through mitosis but also through meiosis. This prompted the idea that perturbations in one generation can be passed on to the next[4]. The all-encompassing epi+genetics definition in combination with the interpretability trap described above were primed for over-interpretation of studies of inter- or trans-generational transmission of molecular regulatory information. It is fair to say that a lot of the early work in this area has been excessively over-interpreted.

The second unintended consequence followed the release of the word epigenetics into the public realm, where it was quickly domesticated by people interested in genetic non-determinism, who enthusiastically stitched together environment, epigenetics and their favoured intervention. This has prompted commercial offerings such as epigenetic yoga, epigenetic face cream and epigenetic anti-dandruff shampoo, all currently available with the click of a mouse.

The solution: say what we mean

Using the term epigenetics on its own can no longer be justified considering its multiple interpretations, so we need to be clear which definition we are using. If we mean epi+genetics — the layer of information beyond the genome — it is unclear why we don't just say 'transcription regulation' instead. If our definition is basically a proxy for the mediation of environmental responses, this should not be equated with epigenetic processes at all. My preference is to reserve the term to describe properties involving cell states (cellular reprogramming) and fates (polycreodism), which are mediated by but not equivalent to the many epi+genetic transcription regulators. Whatever your favoured definition, spell it out clearly to let the audience understand what you mean.

A handy checklist

- Do you want to use the word 'epigenetic' to describe a cellular memory, persistent homeostasis in the absence of the original perturbation or an effect on cell fate not attributable to changes in DNA sequence?

Go ahead. Waddington and Nanney will be pleased with you.

- Do you want to describe a higher level of information that exists beyond the genome and instructs genes how and when to switch on and off?

You are describing transcription regulation. Unless you are also describing the situation in the first point, you should not call this epigenetic, even if you think it makes your research sound sexier. Apply scientific accuracy and use terms like 'transcription regulation'.

- You have found a difference in DNA methylation when comparing two sets of samples, but have not accounted for confounding influences like the proportions of cell subtypes, DNA sequence polymorphism or reverse causation. You want to call this 'epigenetic'?

No. You can't.

- You want to know if it's OK to add royal jelly to petroleum jelly and market it as 'epigenetic face cream'?

If you can sell this to the gullible, go ahead. But we can no longer be friends.

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Competing interests statement
The author declares no competing interests.